

Qi Chen

E-mail: qi.chen-3@manchester.ac.uk, cq019@outlook.com

Tel: +44 7529282060, +86 15165315679

Education

Ph.D. in Fluids and Soft Matter, University of Manchester, 2025

- Thesis: *Microfluidic Model of Haemodynamics in Porous Media*
- Supervisors: Prof. Anne Juel and Dr. Igor Chernyavsky
- Research Focus: microfluidics, haemodynamics, elastic capsules, suspension flow, porous media

M.E. in Power Engineering and Engineering Thermophysics, Shandong University, 2020

- Thesis: *Modelling and Experiment of Droplet Formation in non-Newtonian Fluids*
- Supervisor: Dr. Xuefang Li
- Research Focus: droplet microfluidics, non-Newtonian fluids, bubbles, porous media, hydrogen safety

B.E. in Energy and Power Engineering, Shandong University, 2017

Research Experience

Haemodynamics in human placenta, Ph.D. Researcher, 2021 – present

- Developed a microfluidic setup to robustly fabricate PDMS capsules as a physical model for red blood cells
- Investigated the global rheology and microscopic transport of dense capsule suspensions (blood analogue) in large planar porous media (placenta model)
- Advancing Placental Research: Unlocking the Mysteries of Blood Flow in Pregnancy:
<https://www.youtube.com/watch?v=DXtKvrAFQsQ>

Droplet Microfluidics, Key member, 2018 – 2020

- Set up a microfluidics experimental system and observed the droplet formation in flow-focusing devices
- Modelled the Newtonian droplet formation in non-Newtonian fluids using OpenFOAM
- Enhanced oil recovery using micro bubbles

Hydrogen Safety, Key Member, 2018 – 2020

- Numerically investigated the underexpanded hydrogen jet through square and rectangular nozzles
- Performed experiments of transient and steady jets of high pressure helium facing obstacles
- Designed a miniature of real underground garage and measured the helium concentration distributions when helium leaks to simulate the hydrogen leakages of hydrogen fuel cell vehicles
- Conducting hydrogen leakage simulations and quantitative risk analysis (QRA) in a fuel cell train

Analysis and Optimisation of Heat Exchanger, Participant, 2017 – 2018

- Modelled the fluid flow and heat transfer in continuous helical baffled heat exchangers
 - Designed five elliptical tube layouts and analysed the comprehensive performances
-

Publications

- [1] **Qi Chen**, Naval Singh, Kerstin Schirrmann, Qi Zhou, Igor L. Chernyavsky, and Anne Juel. Robust fabrication of ultra-soft tunable PDMS microcapsules as a biomimetic model for red blood cells. *Soft Matter* 19 (2023): 5249-5261. Doi: [10.1039/D3SM00208J](https://doi.org/10.1039/D3SM00208J).
- [2] Eleanor Doman, Qi Zhou, **Qi Chen**, Naval Singh, Anne Juel, Miguel O. Bernabeu, Timm Krüger, Oliver E. Jensen, and Igor L. Chernyavsky. Characterization of micro-haemodynamics in the placental intervillous space by integrated experimental and theoretical approaches. *Placenta* 140 (2023): E63-E64. Doi: [10.1016/j.placenta.2023.07.205](https://doi.org/10.1016/j.placenta.2023.07.205).
- [3] **Qi Chen**, Eleanor Doman, Naval Singh, Igor Chernyavsky, Oliver Jensen, and Anne Juel. Microfluidic model of micro-haemodynamics in porous media. *Bulletin of the American Physical Society* (2023).
- [4] Qi Zhou, Kerstin Schirrmann, Eleanor Doman, **Qi Chen**, Naval Singh, P. Ravi Selvaganapathy, Miguel O. Bernabeu et al.

- Red blood cell dynamics in extravascular biological tissues modelled as canonical disordered porous media. *Interface Focus* 12, no. 6 (2022): 20220037. Doi: [10.1098/rsfs.2022.0037](https://doi.org/10.1098/rsfs.2022.0037).
- [5] Qi Zhou, Eleanor Doman, Kerstin Schirrmann, **Qi Chen**, Elizabeth A. Seed, Edward D. Johnstone, P. Ravi Selvaganapathy et al. Micro-haemodynamics at the maternal–fetal interface: Experimental, theoretical and clinical perspectives. *Current Opinion in Biomedical Engineering* 22 (2022): 100387. Doi: [10.1016/j.cobme.2022.100387](https://doi.org/10.1016/j.cobme.2022.100387).
- [6] **Qi Chen**, Naval Singh, Kerstin Schirrmann, Igor Chernyavsky, and Anne Juel. Microfluidic model of micro-haemodynamics in complex porous media. *Bulletin of the American Physical Society* 67 (2022).
- [7] **Qi Chen**, Jingkun Li, Yu Song, Bin Chen, David M. Christopher, and Xuefang Li. Pressure-driven microfluidic droplet formation in Newtonian and shear-thinning fluids in glass flow-focusing microchannels. *International Journal of Multiphase Flow* 140 (2021): 103648. [10.1016/j.ijmultiphaseflow.2021.103648](https://doi.org/10.1016/j.ijmultiphaseflow.2021.103648).
- [8] **Qi Chen**, Jingkun Li, Yu Song, David M. Christopher, and Xuefang Li. Modeling of Newtonian droplet formation in power-law non-Newtonian fluids in a flow-focusing device. *Heat and Mass Transfer* 56 (2020): 2711-2723. Doi: [10.1007/s00231-020-02899-6](https://doi.org/10.1007/s00231-020-02899-6).
- [9] **Qi Chen**, Jingkun Li, Yu Song, Qian He, and Xuefang Li. Newtonian droplet generation in shear-thinning fluids in flow-focusing microchannel. *CIESC Journal* 71 (2020):1510.
- [10] Tingting Du, **Qi Chen**, Wenjing Du, and Lin Cheng. Performance of continuous helical baffled heat exchanger with varying elliptical tube layouts. *International Journal of Heat and Mass Transfer* 133 (2019): 1165-1175. Doi: [10.1016/j.ijheatmasstransfer.2018.12.142](https://doi.org/10.1016/j.ijheatmasstransfer.2018.12.142).
- [11] Xuefang Li, **Qi Chen**, Mingjia Chen, Qian He, David M. Christopher, Xu Cheng, Bikram Roy Chowdhury, and Ethan S. Hecht. Modeling of underexpanded hydrogen jets through square and rectangular slot nozzles. *International journal of hydrogen energy* 44, (2019): 6353-6365. Doi: [10.1016/j.ijhydene.2019.01.079](https://doi.org/10.1016/j.ijhydene.2019.01.079).
-

Conference presentations

- **Microfluidic model of micro-haemodynamics in porous media**, *APS Division of Fluid Dynamics Annual Meeting*, Washington USA, November 2023, oral presentation.
 - **A microfluidic model of micro-haemorheology in complex porous media**, *International Congress on Rheology*, Athens Greece, August 2023, oral presentation.
 - **Microfluidic model of micro-haemodynamics**, *APS Division of Fluid Dynamics Annual Meeting*, Indianapolis USA, November 2022, oral presentation.
 - **Microfluidic model of micro-haemodynamics**, *IOP workshop: Microrheology and Transport in Complex Biological Media*, Edinburgh UK, May 2022, oral presentation.
 - **Microfluidic model of micro-haemodynamics**, *Biological Fluids & Flows: A Conference to Celebrate the 80th Birthday of Prof. Tim Pedley*, Cambridge UK, March 2022, poster presentation
 - **A robust microfluidic device for fabricating deformable microcapsules based on water-oil-water double-emulsion templates**, *UK Fluids Conference*, online, 2021, poster presentation
-

Teaching Experience

Teaching Assistant, 1st year lab teaching, University of Manchester, 2022

- Supervised lab sessions and student projects.
-

Technical Skills

- **Programming & Software:** MATLAB, Python, OpenFOAM, LabView, Autodesk Inventor, etc.
- **Experimental Techniques:** Microfluidics, Rheometry, High-speed imaging, Pressure measurement, Hydrogen jet and diffusion, etc.
- **Data Analysis:** Statistical modelling, Image processing, Machine learning, etc.